PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR 2025-2026 | FIFTH SEMESTER B.TECH

UNIT TEST REPORT

(Unit and Integration Test Execution)

PROJECT: SPRINTFLOW – PROJECT MANAGEMENT SYSTEM

(Workflow, Visibility, Estimation, and Kanban)

SUBMITTED BY: TEAM 11 (SECTION 5C)

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Repository: https://github.com/Darshanpawar7/SE-SRS

Document: Unit Test Report, Testing Phase (Part 3, item 2.b.i)

Report owner: Gurubelli Yekambar Eshwar Rao, QA Lead (GitHub issue #91)

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Date: 11-11-2026

Code under test: `main` branch, commit `f9e0727`; final regressions on `main` at commit `2b8fdbc` (06-11-2026) and commit `970c656` (11-11-2026) (Section 10)

Commit IDs: On 11-11-2026 some commit messages in the repository history were edited. The code did not change, but the commit IDs did. This report keeps the IDs recorded during testing; on GitHub today `f9e0727` is `2398c89` and `2b8fdbc` is `21f3681`. GitHub Actions runs from before that date still show the old IDs.

Approved by: Darshan P Pawar, Test Architect (STP Section 9), 11-11-2026

Repository location: `docs/test-reports/UNIT_TEST_REPORT.md` and raw results in `docs/test-reports/evidence/`

Revision History

Version	Date	Author	Change Summary
1.0	06-11-2026	Team 11	Results of the automated unit and integration suite for the testing phase, with per-file results, key test cases, CI history and observations
1.1	11-11-2026	Team 11	Added the pre-submission regression on the final `main` branch (Section 10: 272 of 272 tests pass in GitHub Actions run 42973669659). Corrected details found in a final review: the CI history (Section 9), the source of the benchmark figures (Section 8), the modules three test files load, the requirement mapping (Section 7) and the test inputs in Appendix A. Updated observation OBS-01 and the conclusion, and noted the changed commit IDs. The results of the testing phase are unchanged.

1. Introduction

1.1 Purpose

This report records the unit and integration testing of SprintFlow carried out in the testing phase, 2 to 6 November 2026. Unit testing is the first test level in our Software Test Plan (STP v1.0, Section 5.1), and its result is an entry criterion for system testing: all unit tests must pass before the system is tested end to end. The STP schedule (Section 7) first ran the unit suite on 11-09-2026, during Sprint 1; the course brief set this testing phase for 2 to 6 November, so the much larger Sprint 2 suite was run again in full for this report.

1.2 Scope

The scope is the automated Vitest suite in `src/tests`: 35 test files with 234 test cases. Between them they test:

- the business logic in `src/utils`: estimation, workflow transitions, burndown, audit trail, filtering, sprint rules and the Sprint 2 analytics;
- the data service in `src/services`: offline storage, the Supabase sync queue and the credential checks, with storage and the Supabase client replaced by test doubles;
- the two React contexts and a set of components (Kanban board, task dialog, KPI cards, burndown chart, workload list and sign-in view), rendered to HTML in Node without a browser.

Testing of the running application in real browsers is covered in the System Validation Report. How much of the source code these tests execute is covered in the Code Coverage and Branch Coverage Report.

1.3 References

- Software Test Plan, STP v1.0 (`docs/STP_Document_Team11.md`), Sections 5, 6, 13 and 14
- Software Requirements Specification (`docs/SRS_Document_Team11.md`)
- Test Case Specification (`docs/TEST_CASE_SPECIFICATION.md`)
- Sprint 1 deviations and ID mapping (`docs/SPRINT1_DEVIATIONS.md`, Section 1)
- CI workflow (`.github/workflows/ci.yml`)

2. Test Environment

Item	Value
Code under test	`main` branch at commit `f9e0727` (merge of PR #90, 3 November 2026)
Test framework	Vitest 4.1.12, test environment `node` (configured in `vite.config.js`)
Local machine	Windows 11 Pro, Node.js 24.15.0
CI runner	GitHub Actions, `ubuntu-latest`, Node.js 20
Command	`npm test` (runs `vitest run`)
Test data	Seed data in `src/services/mockData.js` and fixtures written inside each test
External services	None. No test connects to a real Supabase project; tests that need one use a fake client.
Raw results	`docs/test-reports/evidence/unit-test-results.junit.xml` and `unit-test-results.json`

The STP (Section 6.2) lists Vite 5 and Vitest 2. Both were upgraded in Sprint 2 when we fixed vulnerable dependencies (issue #30, PR #76), so the suite now runs on Vite 7 and Vitest 4.

3. Test Approach

Each test file targets one feature. Thirteen of the 35 files name their `describe` block after the SRS requirement they check, for example "PMS-F-014 immutable audit trail". These names use SRS numbers, while the Test Case Specification uses the Test Plan's numbers (mapping in `docs/SPRINT1_DEVIATIONS.md`, Section 1). When a unit test reproduces a case from the Test Case Specification, the case ID is part of the test name (TC-AUTH-01, TC-LOG-01, TC-OFF-01, TC-RECOV-01 and TC-SET-01), so the result can be traced back to the specification.

Kind of test	What it checks	Example files
Pure unit tests	One function from `src/utils`, called with fixed inputs and compared with an expected output	`estimation`, `sprintMetrics`, `taskFilters`, `taskUpdates`
Service (integration) tests	The data service in `src/services` together with the utilities it calls, with storage and the Supabase client replaced by test doubles	`offlineFallback`, `supabaseSync`, `persistenceRecovery`, `auditTrail`
Component tests	A React component rendered to static HTML and checked for text, labels and ARIA roles	`kpiDashboard`, `workload`, `accessibility`, `auth`

Boundary and negative inputs are written with `it.each`, so a single test body runs over a list of bad values, for example 13 invalid story point values and 7 illegal sprint transitions. A test passes only if every assertion in it holds.

4. Execution Summary

Measure	Result
Test files executed	35
`describe` groups	28 (the other 7 files list their tests at the top level)
Test cases planned	234 (every test in the suite)
Test cases executed	234 (execution rate 100%, STP target 100%)
Passed	**234**
Failed	0
Skipped	0
Pass rate	**100%**
Duration	About 2 seconds

Run	Where	Date and time	Result
Recorded run (results in this report)	Local, Windows 11	06-11-2026, 16:57 IST	234 passed, 0 failed
CI run for commit `f9e0727` (run #233)	GitHub Actions	03-11-2026, 23:00 IST	234 passed, 0 failed (Figure 1)
Re-check before submission	Local, Windows 11	06-11-2026, 21:34 IST	234 passed, 0 failed

Verdict: Pass. All 234 tests passed in every run listed above. This meets the STP entry criterion for system testing ("`npm test` reports 0 failures", Section 5.3).

```
 RUN  v4.1.12 /home/runner/work/SE-SRS/SE-SRS

 ✓ src/tests/supabaseSync.test.js (15 tests) 35ms
 ✓ src/tests/auditTrail.test.js (17 tests) 64ms
 ✓ src/tests/offlineFallback.test.js (14 tests) 43ms
 ✓ src/tests/persistenceRecovery.test.js (11 tests) 137ms
 ✓ src/tests/projectMetadata.test.js (20 tests) 38ms
 ✓ src/tests/burndown.test.js (8 tests) 31ms
 ✓ src/tests/taskForm.test.js (5 tests) 8ms
 ✓ src/tests/accessibility.test.jsx (6 tests) 29ms
 ✓ src/tests/auth.test.js (10 tests) 39ms
 ✓ src/tests/workflow.test.js (8 tests) 10ms
 ✓ src/tests/workload.test.js (5 tests) 20ms
 ✓ src/tests/sprintLifecycle.test.js (16 tests) 8ms
 ✓ src/tests/userText.test.js (4 tests) 16ms
 ✓ src/tests/kanbanPerformance.test.jsx (1 test) 606ms
     ✓ records the state update and Kanban component render path for a 100-task fixture  605ms
 ✓ src/tests/labels.test.js (6 tests) 14ms
 ✓ src/tests/sprintReports.test.js (4 tests) 17ms
 ✓ src/tests/kpiDashboard.test.js (4 tests) 14ms
 ✓ src/tests/savedWorkViews.test.js (5 tests) 19ms
 ✓ src/tests/workspaceFeedback.test.js (5 tests) 9ms
 ✓ src/tests/deliverySignals.test.js (6 tests) 9ms
 ✓ src/tests/projectMap.test.js (5 tests) 8ms
 ✓ src/tests/configSecurity.test.js (3 tests) 5ms
 ✓ src/tests/storyPoints.test.js (16 tests) 15ms
 ✓ src/tests/flowAnalytics.test.js (4 tests) 16ms
 ✓ src/tests/taskUpdates.test.js (3 tests) 7ms
 ✓ src/tests/sprintMetrics.test.js (6 tests) 5ms
 ✓ src/tests/estimation.test.js (3 tests) 6ms
 ✓ src/tests/databaseSecurity.test.js (2 tests) 6ms
 ✓ src/tests/taskExplorer.test.js (10 tests) 16ms
 ✓ src/tests/taskFilters.test.js (2 tests) 6ms
 ✓ src/tests/workSignals.test.js (3 tests) 10ms
 ✓ src/tests/workViews.test.js (2 tests) 5ms
 ✓ src/tests/commands.test.js (3 tests) 5ms
 ✓ src/tests/taskSignals.test.js (1 test) 9ms
 ✓ src/tests/recovery.test.js (1 test) 6ms

 Test Files  35 passed (35)
      Tests  234 passed (234)
   Start at  17:31:14
   Duration  3.34s (transform 1.23s, setup 0ms, import 3.27s, tests 1.29s, environment 4ms)
```

Figure 1: Vitest output from the GitHub Actions run for commit f9e0727 (job "Automated Unit & Workflow Tests"). Warnings printed to stderr are left out here and discussed in Section 11.

5. Results by Test File

The modules column lists the application modules each test file imports.

#	Test file	Modules under test	Tests	Pass	Fail	Time
1	`accessibility.test.jsx`	`utils/accessibility`, `components/kanban/KanbanBoard`, `components/kanban/TaskModal`, `context/ProjectContext`, `context/AuthContext`, `utils/constants`	6	6	0	30 ms
2	`auditTrail.test.js`	`utils/auditTrail`, `services/supabaseClient`, `services/mockData`, `components/dashboard/AuditTrail`, `context/AuthContext`, `context/ProjectContext`	17	17	0	178 ms

#	Test file	Modules under test	Tests	Pass	Fail	Time
3	auth.test.js	services/authService, services/supabaseClient, utils/constants, context/AuthContext, context/ProjectContext, components/auth/SignInView, App	10	10	0	55 ms
4	burndown.test.js	utils/burndown, services/supabaseClient, components/dashboard/BurndownChart	8	8	0	45 ms
5	commands.test.js	utils/commands	3	3	0	3 ms
6	configSecurity.test.js	services/supabaseClient	3	3	0	5 ms
7	databaseSecurity.test.js	–	2	2	0	5 ms
8	deliverySignals.test.js	utils/deliverySignals	6	6	0	12 ms
9	estimation.test.js	utils/estimation	3	3	0	5 ms
10	flowAnalytics.test.js	utils/flowAnalytics	4	4	0	23 ms
11	kanbanPerformance.test.jsx	utils/constants, utils/taskUpdates, components/kanban/KanbanBoard, context/ProjectContext, context/AuthContext	1	1	0	939 ms
12	kpiDashboard.test.js	utils/estimation, utils/taskUpdates, utils/kpis, components/dashboard/KpiCards	4	4	0	20 ms
13	labels.test.js	utils/labels, components/kanban/LabelPills, services/supabaseClient	6	6	0	18 ms
14	offlineFallback.test.js	services/supabaseClient, context/AuthContext, context/ProjectContext, components/layout/Navbar, services/mockData, utils/auditTrail, services/authService, utils/constants	14	14	0	93 ms
15	persistenceRecovery.test.js	services/mockData, utils/auditTrail, services/localPersistence, services/supabaseClient	11	11	0	214 ms
16	projectMap.test.js	utils/projectMap	5	5	0	9 ms
17	projectMetadata.test.js	utils/projectMetadata, utils/auditTrail, services/supabaseClient, services/mockData, utils/constants, context/AuthContext, context/ProjectContext, components/settings/ProjectSettingsModal	20	20	0	46 ms
18	recovery.test.js	components/workspace/ErrorBoundary	1	1	0	6 ms
19	savedWorkViews.test.js	utils/taskExplorer, utils/savedWorkViews	5	5	0	27 ms
20	sprintLifecycle.test.js	utils/sprintLifecycle, services/supabaseClient	16	16	0	13 ms
21	sprintMetrics.test.js	utils/estimation	6	6	0	7 ms
22	sprintReports.test.js	utils/sprintReports	4	4	0	22 ms
23	storyPoints.test.js	utils/constants, utils/storyPoints, services/supabaseClient, services/mockData	16	16	0	20 ms
24	supabaseSync.test.js	services/supabaseClient, services/mockData, utils/auditTrail	15	15	0	38 ms
25	taskExplorer.test.js	utils/taskExplorer	10	10	0	20 ms
26	taskFilters.test.js	utils/taskFilters	2	2	0	6 ms
27	taskForm.test.js	services/supabaseClient, utils/constants, utils/taskForm	5	5	0	10 ms
28	taskSignals.test.js	components/workspace/TaskSignals	1	1	0	7 ms

#	Test file	Modules under test	Tests	Pass	Fail	Time
29	`taskUpdates.test.js`	`utils/taskUpdates`	3	3	0	9 ms
30	`userText.test.js`	`utils/taskForm`, `utils/projectMetadata`, `utils/labels`, `utils/userText`, `components/kanban/LabelPills`, `utils/constants`	4	4	0	19 ms
31	`workflow.test.js`	`utils/constants`, `utils/kanban`, `utils/workflow`, `services/supabaseClient`	8	8	0	11 ms
32	`workload.test.js`	`utils/estimation`, `components/dashboard/WorkloadList`	5	5	0	17 ms
33	`workSignals.test.js`	`utils/workSignals`, `utils/taskExplorer`, `utils/auditTrail`	3	3	0	13 ms
34	`workspaceFeedback.test.js`	`utils/workspaceFeedback`, `services/feedbackStore`	5	5	0	12 ms
35	`workViews.test.js`	`utils/workViews`	2	2	0	5 ms
	Total (35 files)		**234**	**234**	**0**	

6. Key Unit Test Cases

The table gives the input and expected result of a representative set of unit tests, taken from the test code. The actual result is what the recorded run produced. The complete list of all 234 test cases is in Appendix A.

ID	Unit under test (test file)	Input	Expected result	Actual result	Status
UT-01	`calculateSprintMetrics` estimation.test.js	Four tasks: 5 pts done, 8 pts in progress, 3 pts to do, 5 pts done	4 tasks, 2 done, 21 total points, 10 completed, 11 remaining, progress 48%	As expected	**Pass**
UT-02	`calculateSprintMetrics` estimation.test.js	Empty task list	0 tasks, 0 points, progress 0% (no division by zero)	As expected	**Pass**
UT-03	Sprint totals for the selected sprint sprintMetrics.test.js	Tasks spread over sprint s1, sprint s2 and the backlog; s1 selected	Only s1 counted: 2 tasks, 10 points, 8 completed, 2 remaining, 80%	As expected	**Pass**
UT-04	`DataService.saveTasks` storyPoints.test.js	A saved 3-point task, then a task with 0, -1, 4, 6, 34, 1.5, NaN, Infinity, null, undefined, empty, the string "5" or true (13 cases)	Save rejected with a "Story points" error; the saved 3-point task is unchanged	As expected	**Pass**
UT-05	`transitionSprint` sprintLifecycle.test.js	Planned → Completed, and 6 other illegal moves (e.g. Completed → Active)	Error: sprints go planned to active to completed	As expected	**Pass**
UT-06	`advanceTaskByOneStage` workflow.test.js	Task in To Do, advanced by Dhatri	Returns true; update called once with status In Progress	As expected	**Pass**
UT-07	`getTaskAfterStatusDrop` taskUpdates.test.js	To Do task dropped on Code Review; dropped on To Do; unknown task id	First drop gives status Code Review; the other two return null (no change)	As expected	**Pass**
UT-08	`validateTaskForm` taskForm.test.js	A title of only spaces with High; "Valid title" with priority "urgent"; "Fix a bug" with spaces around it and High	false; false; true	As expected	**Pass**
UT-09	`buildTaskPayload` userText.test.js	Title `Fix <script>alert(1)</script> safely`, an `<img onerror>` in the description, script tags in labels	Title "Fix safely" and description "Details remain readable" (tags removed, text kept), labels [security, urgent]	As expected	**Pass**
UT-10	`validateProjectMetadata` projectMetadata.test.js	Project key "", "P", "1PMS", "PM-S", "PMS 2" or "ABCDEFGHIJK"	Each rejected with "Project key must be ..."	As expected	**Pass**
UT-11	`signIn` auth.test.js	Lead email with a wrong password; a blank password; intruder@example.com	"Incorrect email or password"; "Enter your password"; "Only Team 11 accounts"; no session stored	As expected	**Pass**

ID	Unit under test (test file)	Input	Expected result	Actual result	Status
UT-12	`isValidSupabaseAnonKey` configSecurity.test.js	A JWT whose role is service_role; a JWT whose role is anon	false (service key refused); true	As expected	**Pass**
UT-13	`saveSupabaseConfig` (TC-SET-01) supabaseSync.test.js	Valid config saved first, then "not a url", "http://team11.supabase.co" and a blank key	All three rejected; the saved https config is kept	As expected	**Pass**
UT-14	Audit trail through `DataService` (TC-LOG-01) auditTrail.test.js	Create a task, rename it, delete it	Exactly 3 new records, CREATE, UPDATE, DELETE in order, each with actor, task id and timestamp; UPDATE details show old → new title	As expected	**Pass**
UT-15	Offline mode (TC-OFF-01) offlineFallback.test.js	No Supabase client; save tasks 1 and 2, mark 1 done, delete 2, reload	Only task 1 (done) is loaded back; 0 changes waiting to sync	As expected	**Pass**
UT-16	`generateRecordedBurndown` burndown.test.js	One-day sprint; date before the sprint starts; null sprint; end date "bad"	1 point; every actual value null; empty list; empty list (no NaN, no crash)	As expected	**Pass**
UT-17	`csvCell` (CSV export) taskExplorer.test.js	"=1+1", "+cmd", "-cmd", "@SUM(1)", and "=1" / "=cmd" after leading spaces or a tab	Each exported value carries an apostrophe guard, so a spreadsheet shows it as text (the test checks that the apostrophe is present)	As expected	**Pass**
UT-18	`KpiCards` component kpiDashboard.test.js	Metrics of a sprint with no tasks	Rendered HTML shows "0%" and "0 tasks in active scope"; no "NaN"	As expected	**Pass**

7. Requirement Coverage at Unit Level

Requirement IDs follow the SRS, which is our requirements baseline. The Test Plan numbers some features differently, so both IDs are shown. For functional requirements the mapping is in `docs/SPRINT1_DEVIATIONS.md`, Section 1; for the non-functional and security rows, the Test Plan requirement with the same subject is shown.

Feature	SRS ID	Test Plan ID	Unit test files
Sign-in and profile switching	PMS-F-001	PMS-F-001	`auth`
Project metadata	PMS-F-002	–	`projectMetadata`
Five Kanban columns, drag and drop, one-click advance	PMS-F-003 to 005	PMS-F-003 to 005	`workflow`, `taskUpdates`, `kanbanPerformance`, `accessibility`
Create and update tasks (incl. priority)	PMS-F-006	PMS-F-002, PMS-F-016	`taskForm`, `taskUpdates`, `userText`
Labels and tags	PMS-F-007	–	`labels`
Fibonacci story points	PMS-F-008	PMS-F-006	`storyPoints`, `estimation`
Planned points and completion	PMS-F-009	PMS-F-008	`sprintMetrics`, `estimation`
Sprint lifecycle	PMS-F-010	PMS-F-007	`sprintLifecycle`
Burndown chart	PMS-F-011	PMS-F-009	`burndown`
KPI dashboard	PMS-F-012	PMS-F-010	`kpiDashboard`, `deliverySignals`
Team workload	PMS-F-013	PMS-F-017	`workload`
Audit trail	PMS-F-014	PMS-F-012	`auditTrail`
Supabase synchronisation and settings	PMS-F-015	PMS-F-013, PMS-F-015	`supabaseSync`, `configSecurity`
Offline mode	PMS-F-016	PMS-F-014	`offlineFallback`, `persistenceRecovery`
Filtering	PMS-F-017	PMS-F-011	`taskFilters`, `taskExplorer`, `workViews`, `savedWorkViews`
CI/CD pipeline	PMS-F-018	–	Enforced by the workflow itself (Section 9)
In-app documentation viewer	not in SRS	PMS-F-018	None; static content, verified by TC-DOCS-01

Feature	SRS ID	Test Plan ID	Unit test files
Performance under 50 ms	PMS-NF-001	PMS-NF-001	`kanbanPerformance`
Zero data loss across reloads	PMS-NF-004	PMS-NF-006 (closest: no data loss after a network disconnection)	`persistenceRecovery`, `offlineFallback`, `recovery` (its recovery actions keep saved data)
Row Level Security policies	PMS-SR-001	–	`databaseSecurity`
No hardcoded credentials; anon key only	PMS-SR-002	–	`configSecurity`
HTTPS for cloud traffic	PMS-SR-003	PMS-NF-004 (TLS 1.3)	`configSecurity`, `supabaseSync`
Input sanitisation (XSS)	PMS-SR-004	–	`userText`
Sprint 2 enhancements	–	–	`projectMap`, `flowAnalytics`, `sprintReports`, `workSignals`, `taskSignals`, `workspaceFeedback`, `commands`

Every SRS functional requirement has at least one unit test file except PMS-F-018 (the CI/CD pipeline), which is the workflow that runs these tests on every push.

STP Section 5.1 also names three integration pairs. AuthContext with SupabaseClient is exercised in `auth`, `auditTrail`, `offlineFallback` and `projectMetadata`, and ProjectContext with KanbanBoard in `accessibility` and `kanbanPerformance`. SprintManager with Estimation has no unit test: the estimation logic is unit-tested on its own, but no unit test renders SprintManager. That pair is covered at system level by TC-SPR-01, TC-EST-01 and TC-EST-02 (System Validation Report).

8. Performance Benchmark (Unit Level)

`kanbanPerformance.test.jsx` times the drag-and-drop state update and the Kanban re-render for a 100-task fixture (5 warm-up runs, then 30 measured samples) against the 50 ms budget of PMS-NF-001.

Measure	p50	p95	Max
State update (`applyTaskUpdate`)	0.01 ms	0.03 ms	–
Board render (`renderToStaticMarkup`)	7.6 ms	13.6 ms	–
Combined	7.6 ms	**13.7 ms**	17.2 ms

These figures come from a standalone run of the benchmark on 06-11-2026 at 16:59 IST, just after the recorded run; the recorded run's JSON results do not include the benchmark's console output. The GitHub Actions run for commit `f9e0727` (Figure 1) measured a p50 of 14.5 ms, a p95 of 21.4 ms and a maximum of 25.8 ms. Result: pass in both. The 95th percentile is 13.7 ms locally and 21.4 ms on GitHub Actions, well under the 50 ms budget.

This test measures wall-clock time, so its margin depends on the machine it runs on. On 6 November we ran it 10 times on its own and it passed all 10 times. Inside full parallel runs on the same laptop, while OneDrive sync, Docker and a browser were also busy, its 95th percentile rose to as much as 78 ms and the test failed. It has passed in every completed CI run on `main` since 3 November. We record this as observation OBS-01 in Section 11; it is a limitation of the test, not slow code.

9. Continuous Integration History

The same suite runs on every push and pull request in the "CI/CD Pipeline - Team 11" GitHub Actions workflow, together with the build and lint. A separate "Dependency Security Audit" workflow runs `npm audit`.

Scope	Runs	Passed	Failed	Other
`main`, CI/CD Pipeline, since 5 Sep 2026	50	41	7	2 cancelled
`main`, Dependency Security Audit, since 2 Nov 2026	10	10	0	0
All branches, last 200 workflow runs	200	188 (94.0%)	10	2

Five of the seven failures on `main` happened on 2 November, when a merge left conflict markers in `TaskModal.jsx` and the build broke. PR #84 fixed it on 3 November, and every completed `main` run since then has passed. Two runs were cancelled because a newer merge replaced them. The other two failures date from 15 September.

The STP target for the CI pass rate is 95% (Section 14.1). Over the last 200 runs on all branches the rate is 94.0%, just under the target. All ten failures in that window happened on 2 November: the five on `main` described above, and five on two feature branches the same day.

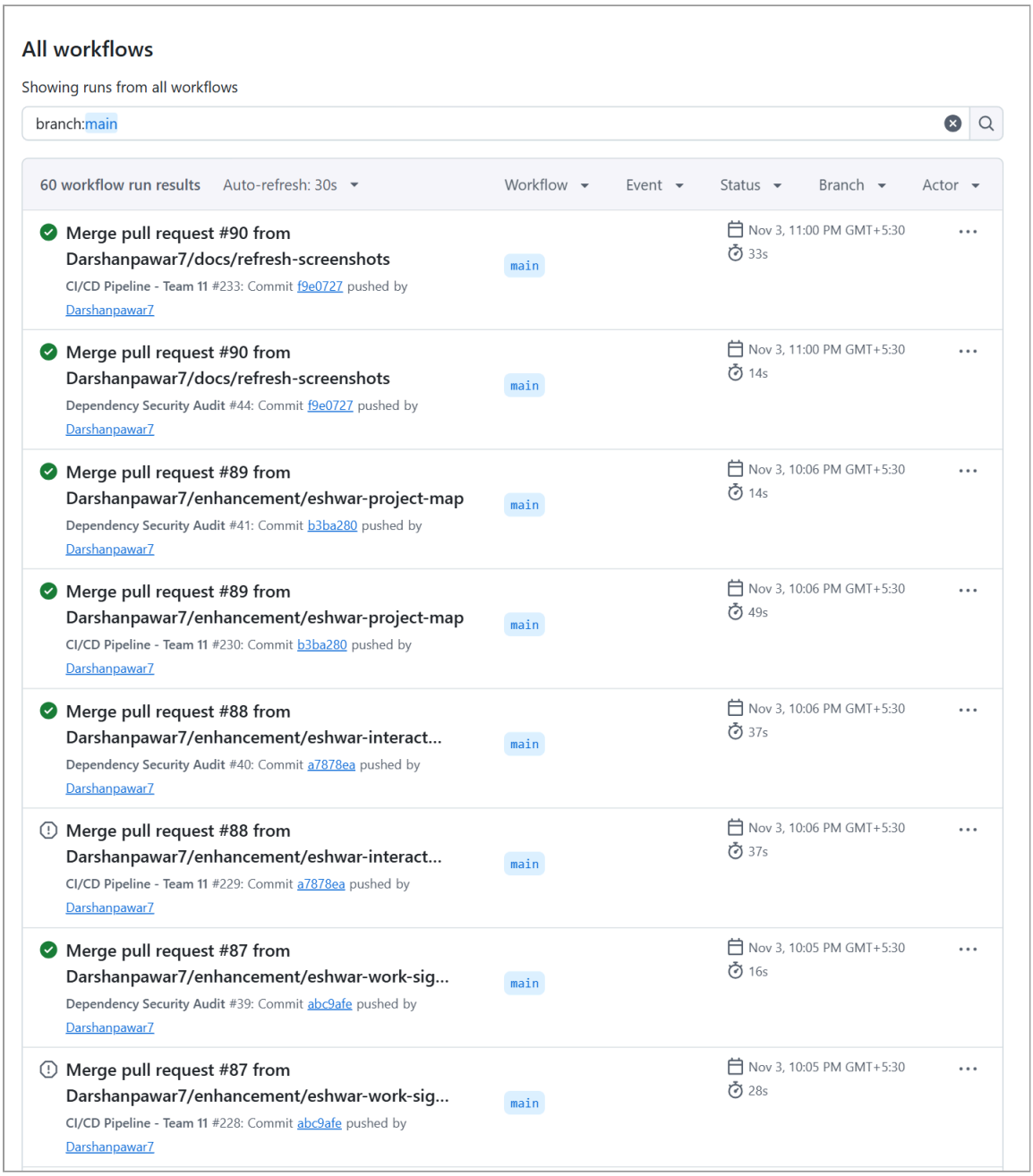


Figure 2: GitHub Actions runs on the main branch (public repository page, captured 06-11-2026). The two runs marked "!" were cancelled by a newer merge.

10. Regression on the Final main Branch

After the recorded run, five pull requests were merged into `main` on 06-11-2026: #97 (the first version of these reports, which also added the coverage tooling and the system test script without changing the application code or the unit tests), #98 (the fix for system test defect DEF-04, with new tests), #99 (a second-machine pre-check of the system tests), #101 (the fix for DEF-05 and for observations OBS-02 and OBS-03 below) and #85 (verified cloud sessions, merged after our review fixes, with 21 new tests). We ran the whole suite again on the final `main` that evening.

Four more pull requests were merged before submission: #103 (completion metrics and bounded flow history), #104 (task sync that still works with the earlier database schema), #105 (save feedback, sign-in appearance and project map performance) and #107 (automated test evidence and accessibility fixes). On 11-11-2026 the "Sprint 1-2 Testing Evidence" workflow ran the whole suite again on the final `main` (GitHub Actions run 42973669659, Ubuntu 24.04, Node.js 24.22.2).

Run	Commit	Test files	Tests	Passed	Failed
Recorded run (Sections 4 to 7)	`f9e0727`	35	234	234	0
Final regression, 06-11-2026, 23:04 IST	`2b8fdbc`	38	259	259	0
Pre-submission regression, 11-11-2026, 12:07 IST (GitHub Actions)	`970c656`	41	272	272	0

All 272 tests pass on the final `main`. The 25 tests added on 06-11-2026 cover the board's empty filter state (3), the "1 task" wording (1), and cloud session verification and cache ownership (21); lint, the production build and `npm audit --audit-level=high` were clean on that commit, and the GitHub Actions pipeline, now on Node.js 24, passed for every pull request merged that day. The 13 tests added since then are in three new files: `analyticsReview.test.js` (6 tests, PR #103), `schemaCompatibility.test.js` (4, PR #104) and `uiReview.test.js` (3, PR #105). On commit `970c656` the CI pipeline's lint, build and unit test jobs passed (run 42973669847), and so did the dependency audit (run 42973669641).

11. Defects and Observations

The unit tests found no defects in the application code: all 234 tests passed. The build break of 2 November (merge conflict markers in `TaskModal.jsx`, Section 9) was caught by CI and fixed by PR #84 before the recorded run. We recorded three observations about the tests themselves. None of them is a defect in SprintFlow; two have since been fixed and the third partly addressed.

ID	Observation	Impact	Recommendation	Status
OBS-01	`kanbanPerformance.test.jsx` asserts a wall-clock budget, so it fails under coverage instrumentation and can fail on a heavily loaded machine.	Coverage runs and busy laptops report one failure even though the code is not slower.	Skip only the timing assertion when coverage is on, or move the benchmark to a separate `npm run bench` step.	Partly addressed in PR #107: the coverage run now leaves this test out, and the normal run still includes it. It can still fail on a heavily loaded machine.
OBS-02	The CI runner uses Node.js 20, and `@supabase/supabase-js` prints a warning in every run that Node.js 20 and below are deprecated.	None today. A future library update may stop working on the CI runner.	Move the CI pipeline to a supported Node.js version.	Fixed in PR #101: CI runs on Node.js 24 and the warning is gone
OBS-03	While rendering `BurndownChart`, React warns that an SVG `<title>` element "received an array with more than 1 element as children".	The tooltip on each burndown point is built from several text pieces; some browsers may not show it as one string.	Build the tooltip as one template string.	Fixed in PR #101: the warning is gone

12. Conclusion

The unit and integration suite is in good shape. All 234 tests pass, in about two seconds locally and about three on GitHub Actions, and every SRS functional requirement except the CI pipeline itself has unit tests behind it. On the same commit, lint reports 0 errors and 0 warnings and `npm audit --audit-level=high` reports no high or critical vulnerabilities. The unit-level evidence for the testing phase is complete, and the entry criterion for system testing is met. On the final `main` all 272 tests pass (GitHub Actions run 42973669659, 11-11-2026; Section 10).

Appendix A: All Unit Test Cases

Every test case from the recorded run, grouped by file. Test names are printed as Vitest reports them: describe block › test name. Where a name hides its input (a blank value, a tab shown as ⇥, or the text "5"), the input is given in brackets.

A.1 `accessibility.test.jsx` (6 tests, 6 passed)

#	Test case	Result	Time
1	keyboard and screen-reader task workflow › opens a focused task card on "Enter" and prevents the default action	Pass	3.2 ms
2	keyboard and screen-reader task workflow › opens a focused task card on " " and prevents the default action	Pass	0.6 ms
3	keyboard and screen-reader task workflow › does not activate for other keys or events bubbling from a child	Pass	1.0 ms
4	keyboard and screen-reader task workflow › renders focusable named task controls without nesting the quick action inside the card button role	Pass	16.0 ms
5	keyboard and screen-reader task workflow › associates task form labels, names the dialog, and announces validation errors	Pass	5.4 ms
6	keyboard and screen-reader task workflow › gives search and filter controls accessible names	Pass	3.1 ms

A.2 `auditTrail.test.js` (17 tests, 17 passed)

#	Test case	Result	Time
1	PMS-F-014 immutable audit trail › creates frozen records with actor, action, target and timestamp	Pass	10.4 ms
2	PMS-F-014 immutable audit trail › rejects records that could not be attributed or dated	Pass	1.3 ms
3	PMS-F-014 immutable audit trail › TC-LOG-01: create, title edit and delete add exactly three records in order	Pass	4.0 ms
4	PMS-F-014 immutable audit trail › TC-LOG-01 through the app: creating, editing and deleting a task each add one record	Pass	9.4 ms
5	PMS-F-014 immutable audit trail › records sprint creation and start made through the app	Pass	2.6 ms
6	PMS-F-014 immutable audit trail › audits sprint deletion and the task changes caused by detaching its tasks	Pass	3.4 ms
7	PMS-F-014 immutable audit trail › keeps long before and after values in full	Pass	0.6 ms
8	PMS-F-014 immutable audit trail › records a plain move as "moved task" and ignores a save with no changes	Pass	0.6 ms
9	PMS-F-014 immutable audit trail › records sprint creation and lifecycle transitions	Pass	0.8 ms
10	PMS-F-014 immutable audit trail › never truncates history and leaves earlier records unchanged	Pass	6.2 ms
11	PMS-F-014 immutable audit trail › keeps both records when two are logged at the same moment	Pass	1.1 ms
12	PMS-F-014 immutable audit trail › keeps the audit trail when demo data is reset, and records the reset	Pass	1.0 ms
13	PMS-F-014 immutable audit trail › only accepts complete audit records	Pass	1.6 ms
14	PMS-F-014 immutable audit trail › formats record times relative to now, falling back to the old display text	Pass	123.7 ms
15	PMS-F-014 immutable audit trail › shows the latest seven records, the total count and a full-history control	Pass	6.6 ms
16	PMS-F-014 immutable audit trail › shows status labels for moves in the seed history	Pass	2.5 ms
17	PMS-F-014 immutable audit trail › records a description-only project change	Pass	0.7 ms

A.3 `auth.test.js` (10 tests, 10 passed)

#	Test case	Result	Time
1	PMS-F-001 team sign-in and profile switching › TC-AUTH-01: signs in a team account and keeps the session after a reload	Pass	7.1 ms
2	PMS-F-001 team sign-in and profile switching › rejects a wrong password, an empty password and accounts outside the team	Pass	3.5 ms
3	PMS-F-001 team sign-in and profile switching › switches profile by signing out and signing in as another member	Pass	1.0 ms

#	Test case	Result	Time
4	PMS-F-001 team sign-in and profile switching › discards a stored session that was edited or corrupted	Pass	1.0 ms
5	PMS-F-001 team sign-in and profile switching › uses Supabase Auth when configured and still rejects non-team accounts	Pass	2.1 ms
6	PMS-F-001 team sign-in and profile switching › only lets the team lead use lead-only actions	Pass	0.8 ms
7	PMS-F-001 team sign-in and profile switching › renders the sign-in view with all four team profiles	Pass	13.3 ms
8	PMS-F-001 team sign-in and profile switching › shows only the sign-in view until a member signs in, then the application	Pass	14.9 ms
9	PMS-F-001 team sign-in and profile switching › offers local demo mode when the Supabase connection was saved in this browser	Pass	3.6 ms
10	PMS-F-001 team sign-in and profile switching › lets only the team lead reset the demo data, and only while Supabase is not connected	Pass	7.1 ms

A.4 `burndown.test.js` (8 tests, 8 passed)

#	Test case	Result	Time
1	PMS-F-011 recorded daily burndown › records actual observations, keeps past days unchanged and leaves gaps/future days empty	Pass	18.6 ms
2	PMS-F-011 recorded daily burndown › keeps only the latest observation today, tracks scope changes, and fixes the initial ideal baseline	Pass	1.4 ms
3	PMS-F-011 recorded daily burndown › isolates sprints and records zero after removal or completion	Pass	0.6 ms
4	PMS-F-011 recorded daily burndown › freezes completed history and records late completion on its real date	Pass	1.9 ms
5	PMS-F-011 recorded daily burndown › does not invent historical values for a legacy sprint first opened after its end	Pass	0.7 ms
6	PMS-F-011 recorded daily burndown › handles empty, future, one-day and invalid sprints without NaN or crashes	Pass	1.2 ms
7	PMS-F-011 recorded daily burndown › persists observations across service reads instead of truncating them to the audit feed	Pass	2.3 ms
8	PMS-F-011 recorded daily burndown › renders exact daily labels and actual values with accessible empty/one-point states	Pass	17.8 ms

A.5 `commands.test.js` (3 tests, 3 passed)

#	Test case	Result	Time
1	Command center › shows actions without flooding the default menu with tasks	Pass	2.1 ms
2	Command center › searches descriptions and titles case-insensitively	Pass	0.3 ms
3	Command center › returns navigation and task matches and caps task results	Pass	0.9 ms

A.6 `configSecurity.test.js` (3 tests, 3 passed)

#	Test case	Result	Time
1	Supabase credential configuration › ignores local environment files and keeps the example credential-free	Pass	3.4 ms
2	Supabase credential configuration › accepts only HTTPS URLs and non-secret public key values	Pass	1.4 ms
3	Supabase credential configuration › rejects service-role JWTs while allowing legacy anon public JWTs	Pass	0.5 ms

A.7 `databaseSecurity.test.js` (2 tests, 2 passed)

#	Test case	Result	Time
1	Supabase database security schema › enables RLS on every application table and scopes policies to authenticated users	Pass	4.7 ms
2	Supabase database security schema › makes audit records append-only and prevents project deletion from cascading into history	Pass	0.9 ms

A.8 `deliverySignals.test.js` (6 tests, 6 passed)

#	Test case	Result	Time
1	Delivery signals › computes counts from unfinished scoped items and independent review totals	Pass	8.1 ms
2	Delivery signals › provides honest empty and unknown states	Pass	1.0 ms
3	Delivery signals › uses creation fallback and seven elapsed days boundary	Pass	0.6 ms
4	Delivery signals › does not silently count unknown/future dates as stale	Pass	0.6 ms
5	Delivery signals › counts inclusive UTC calendar dates and clamps before/after schedule	Pass	0.7 ms
6	Delivery signals › handles one day and invalid ranges	Pass	0.5 ms

A.9 `estimation.test.js` (3 tests, 3 passed)

#	Test case	Result	Time
1	Agile Estimation & Metric Utilities › calculates sprint metrics correctly	Pass	3.0 ms
2	Agile Estimation & Metric Utilities › handles empty task lists safely without division by zero	Pass	0.5 ms
3	Agile Estimation & Metric Utilities › calculates member workload and completion rates	Pass	0.7 ms

A.10 `flowAnalytics.test.js` (4 tests, 4 passed)

#	Test case	Result	Time
1	recorded flow analytics › uses paired real transitions, deduplicates events, and measures cycles	Pass	18.4 ms
2	recorded flow analytics › does not invent timing for missing starts or future/corrupt observations	Pass	1.7 ms
3	recorded flow analytics › counts reopened completions separately and restricts to current scope	Pass	0.8 ms
4	recorded flow analytics › replaces only today, preserves prior observations and leaves missing dates absent	Pass	2.3 ms

A.11 `kanbanPerformance.test.jsx` (1 test, 1 passed)

#	Test case	Result	Time
1	repeatable local Kanban drag/drop and render benchmark › records the state update and Kanban component render path for a 100-task fixture	Pass	939.2 ms

A.12 `kpiDashboard.test.js` (4 tests, 4 passed)

#	Test case	Result	Time
1	PMS-F-012 executive KPI dashboard › reports total points, delivered velocity, completion and remaining work from the task data	Pass	4.6 ms
2	PMS-F-012 executive KPI dashboard › updates every KPI after a task mutation, with no reload	Pass	2.6 ms
3	PMS-F-012 executive KPI dashboard › renders the four cards with the computed values and an accessible progress bar	Pass	10.4 ms
4	PMS-F-012 executive KPI dashboard › shows zeros, not NaN, when the selected sprint has no tasks	Pass	1.7 ms

A.13 `labels.test.js` (6 tests, 6 passed)

#	Test case	Result	Time
1	PMS-F-007 task labels and tags › normalises typed labels into short lowercase tags	Pass	5.4 ms
2	PMS-F-007 task labels and tags › adds a label once and ignores blank or duplicate input	Pass	2.0 ms
3	PMS-F-007 task labels and tags › removes only the chosen label	Pass	0.7 ms
4	PMS-F-007 task labels and tags › cleans legacy label lists that contain duplicates or non-text values	Pass	0.5 ms
5	PMS-F-007 task labels and tags › renders one distinct pill per saved label on the task card	Pass	7.0 ms
6	PMS-F-007 task labels and tags › keeps added and removed labels across a save and reload	Pass	2.0 ms

A.14 `offlineFallback.test.js` (14 tests, 14 passed)

#	Test case	Result	Time
1	PMS-F-016 offline CRUD fallback › TC-OFF-01: without credentials, create, update and delete persist in local storage across reloads	Pass	8.1 ms
2	PMS-F-016 offline CRUD fallback › keeps sprint create, update and delete changes in local storage across reloads	Pass	5.3 ms
3	PMS-F-016 offline CRUD fallback › queues a sprint deletion and detached tasks until the cloud is reachable	Pass	5.2 ms
4	PMS-F-016 offline CRUD fallback › keeps a change locally and queues it when the cloud save fails	Pass	2.0 ms
5	PMS-F-016 offline CRUD fallback › shows the local change after a reload while the cloud is still unreachable, then syncs it	Pass	2.2 ms
6	PMS-F-016 offline CRUD fallback › TC-RECOV-01: tasks created while disconnected are not lost after reconnecting	Pass	1.5 ms
7	PMS-F-016 offline CRUD fallback › sends an offline delete later, so the deleted task does not come back	Pass	2.2 ms
8	PMS-F-016 offline CRUD fallback › queues audit records that fail to upload and inserts them later	Pass	1.6 ms
9	PMS-F-016 offline CRUD fallback › runs one retry at a time, so a queued record is sent once	Pass	2.0 ms
10	PMS-F-016 offline CRUD fallback › marks a change before sending it, so a reload during a slow save keeps the local copy	Pass	8.5 ms
11	PMS-F-016 offline CRUD fallback › loads the local copy, not the older cloud row, when writes fail but reads still work	Pass	1.7 ms
12	PMS-F-016 offline CRUD fallback › does not let an older save that finishes late clear a newer failed one	Pass	13.3 ms
13	PMS-F-016 offline CRUD fallback › shows waiting changes in the navbar instead of "Supabase Connected"	Pass	35.4 ms
14	PMS-F-016 offline CRUD fallback › keeps every queued change when a retry fails again	Pass	4.5 ms

A.15 `persistenceRecovery.test.js` (11 tests, 11 passed)

#	Test case	Result	Time
1	PMS-NF-004 fresh-session recovery › restores all saved entity types, activity, selection and burndown after a new module session	Pass	159.7 ms
2	PMS-NF-004 fresh-session recovery › queues the audit event before an unanswered entity upload and restores it in a fresh session	Pass	11.8 ms
3	PMS-NF-004 fresh-session recovery › keeps all configured-mode entities through failed uploads, restart and retry	Pass	11.7 ms
4	PMS-NF-004 fresh-session recovery › does not let an older projects read overwrite a newer save	Pass	5.4 ms
5	PMS-NF-004 fresh-session recovery › does not let an older sprints read overwrite a newer save	Pass	3.7 ms
6	PMS-NF-004 fresh-session recovery › does not let an older tasks read overwrite a newer save	Pass	4.0 ms
7	PMS-NF-004 fresh-session recovery › replays an interrupted multi-key write before exposing saved state	Pass	4.0 ms
8	PMS-NF-004 fresh-session recovery › preserves the journal if applying it is interrupted, then replays it	Pass	2.6 ms
9	PMS-NF-004 fresh-session recovery › does not overwrite existing entities when the initial journal cannot be saved	Pass	1.1 ms
10	PMS-NF-004 fresh-session recovery › keeps an audit event saved while an older shared-history read is pending	Pass	3.7 ms
11	PMS-NF-004 fresh-session recovery › recovers sprint deletion, detached tasks, audit and delete retry intent together	Pass	6.0 ms

A.16 `projectMap.test.js` (5 tests, 5 passed)

#	Test case	Result	Time
1	never includes foreign project tasks or sprints in the map	Pass	4.5 ms
2	distinguishes unscheduled work from a missing or foreign sprint	Pass	1.1 ms
3	searches real labels and owners while preserving membership	Pass	1.0 ms
4	limits canvas nodes while keeping every matching task available to the table	Pass	2.3 ms
5	bounds zoom and recovers non-finite values	Pass	0.5 ms

A.17 `projectMetadata.test.js` (20 tests, 20 passed)

#	Test case	Result	Time
1	PMS-F-002 project metadata › accepts and normalises a valid key, name and date range	Pass	7.1 ms
2	PMS-F-002 project metadata › rejects the invalid key ""	Pass	1.9 ms
3	PMS-F-002 project metadata › rejects the invalid key "P"	Pass	0.8 ms
4	PMS-F-002 project metadata › rejects the invalid key "1PMS"	Pass	0.7 ms
5	PMS-F-002 project metadata › rejects the invalid key "PM-S"	Pass	0.6 ms
6	PMS-F-002 project metadata › rejects the invalid key "PMS 2"	Pass	0.5 ms
7	PMS-F-002 project metadata › rejects the invalid key "ABCDEFGHIJK"	Pass	0.5 ms
8	PMS-F-002 project metadata › rejects a key already used by another project, ignoring case	Pass	0.7 ms
9	PMS-F-002 project metadata › rejects invalid details {"name":"AB"}	Pass	0.9 ms
10	PMS-F-002 project metadata › rejects invalid details {"name":"xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx"}	Pass	0.9 ms
11	PMS-F-002 project metadata › rejects invalid details {"start_date":"2026-02-30"}	Pass	0.6 ms
12	PMS-F-002 project metadata › rejects invalid details {"target_date":"30/10/2026"}	Pass	0.4 ms
13	PMS-F-002 project metadata › rejects invalid details {"start_date":"2026-11-01"}	Pass	0.4 ms
14	PMS-F-002 project metadata › saves the details and shows the same values after a reload	Pass	1.7 ms
15	PMS-F-002 project metadata › checks key uniqueness against every Supabase project and syncs the new dates	Pass	3.3 ms
16	PMS-F-002 project metadata › rejects edits from members other than the lead, and edits before the project has loaded	Pass	7.6 ms
17	PMS-F-002 project metadata › stores new details and their audit record before Supabase answers	Pass	7.6 ms
18	PMS-F-002 project metadata › records which project fields changed in the audit trail	Pass	0.7 ms
19	PMS-F-002 project metadata › lets only the team lead edit; other members see the details read-only	Pass	7.8 ms
20	PMS-F-002 project metadata › falls back to placeholder text before the project has loaded	Pass	0.6 ms

A.18 `recovery.test.js` (1 test, 1 passed)

#	Test case	Result	Time
1	Workspace recovery screen › offers non-destructive retry and reload without exposing exception details	Pass	5.7 ms

A.19 `savedWorkViews.test.js` (5 tests, 5 passed)

#	Test case	Result	Time
1	separates saved views by endpoint/account, member and project without key collisions	Pass	3.6 ms
2	recovers corrupt saved data and validates persisted filters against current enums	Pass	3.7 ms
3	bounds saved view count and text and handles duplicate stored identities	Pass	3.1 ms
4	intersects inclusive due-date filters and leaves undated tasks out of a dated range	Pass	15.5 ms
5	orders recorded timestamps with missing dates last and supports explicit unassigned work	Pass	1.0 ms

A.20 `sprintLifecycle.test.js` (16 tests, 16 passed)

#	Test case	Result	Time
1	PMS-F-010 sprint lifecycle › persists planned → active → completed and selected sprint	Pass	5.0 ms
2	PMS-F-010 sprint lifecycle › rejects planned → completed	Pass	1.2 ms
3	PMS-F-010 sprint lifecycle › rejects planned → planned	Pass	0.3 ms
4	PMS-F-010 sprint lifecycle › rejects active → planned	Pass	0.3 ms
5	PMS-F-010 sprint lifecycle › rejects active → active	Pass	0.3 ms
6	PMS-F-010 sprint lifecycle › rejects completed → active	Pass	0.2 ms

#	Test case	Result	Time
7	PMS-F-010 sprint lifecycle › rejects completed → completed	Pass	0.5 ms
8	PMS-F-010 sprint lifecycle › rejects planned → unknown	Pass	0.4 ms
9	PMS-F-010 sprint lifecycle › rejects missing sprints and multiple active sprints in the same project	Pass	1.1 ms
10	PMS-F-010 sprint lifecycle › rejects invalid sprint metadata {"name":" "} *(input: a name of three spaces)*	Pass	0.8 ms
11	PMS-F-010 sprint lifecycle › rejects invalid sprint metadata {"start_date":"invalid"}	Pass	0.6 ms
12	PMS-F-010 sprint lifecycle › rejects invalid sprint metadata {"start_date":"2026-02-30"}	Pass	0.4 ms
13	PMS-F-010 sprint lifecycle › rejects invalid sprint metadata {"end_date":"2026-09-01"}	Pass	0.6 ms
14	PMS-F-010 sprint lifecycle › rejects invalid sprint metadata {"end_date":""}	Pass	0.4 ms
15	PMS-F-010 sprint lifecycle › accepts a same-day sprint and trims its name	Pass	0.3 ms
16	PMS-F-010 sprint lifecycle › restores a valid selection, otherwise prefers an active sprint, then first, then empty	Pass	0.4 ms

A.21 `sprintMetrics.test.js` (6 tests, 6 passed)

#	Test case	Result	Time
1	PMS-F-009 selected sprint totals › uses points, not task counts or other sprint/backlog points	Pass	3.9 ms
2	PMS-F-009 selected sprint totals › returns zero metrics for empty scope null	Pass	0.4 ms
3	PMS-F-009 selected sprint totals › returns zero metrics for empty scope undefined	Pass	0.3 ms
4	PMS-F-009 selected sprint totals › returns zero metrics for empty scope *(input: an empty string)*	Pass	0.2 ms
5	PMS-F-009 selected sprint totals › returns zero metrics for empty scope missing	Pass	0.2 ms
6	PMS-F-009 selected sprint totals › recalculates after completion, estimation, reassignment and deletion	Pass	0.5 ms

A.22 `sprintReports.test.js` (4 tests, 4 passed)

#	Test case	Result	Time
1	validates bounded inclusive UTC ranges, including real calendar dates	Pass	4.2 ms
2	reconciles snapshot totals including unknown statuses, priorities and owners	Pass	2.4 ms
3	exports recorded transitions in the range and excludes duplicates, other tasks and future events	Pass	14.2 ms
4	drills into unique current tasks even when a task has multiple completion transitions	Pass	1.4 ms

A.23 `storyPoints.test.js` (16 tests, 16 passed)

#	Test case	Result	Time
1	PMS-F-008 Fibonacci estimates › offers exactly the SRS scale	Pass	4.4 ms
2	PMS-F-008 Fibonacci estimates › rejects invalid estimate 0 before replacing saved tasks	Pass	3.6 ms
3	PMS-F-008 Fibonacci estimates › rejects invalid estimate -1 before replacing saved tasks	Pass	0.8 ms
4	PMS-F-008 Fibonacci estimates › rejects invalid estimate 4 before replacing saved tasks	Pass	0.7 ms
5	PMS-F-008 Fibonacci estimates › rejects invalid estimate 6 before replacing saved tasks	Pass	0.5 ms
6	PMS-F-008 Fibonacci estimates › rejects invalid estimate 34 before replacing saved tasks	Pass	0.5 ms
7	PMS-F-008 Fibonacci estimates › rejects invalid estimate 1.5 before replacing saved tasks	Pass	0.6 ms
8	PMS-F-008 Fibonacci estimates › rejects invalid estimate NaN before replacing saved tasks	Pass	0.5 ms
9	PMS-F-008 Fibonacci estimates › rejects invalid estimate Infinity before replacing saved tasks	Pass	0.5 ms
10	PMS-F-008 Fibonacci estimates › rejects invalid estimate null before replacing saved tasks	Pass	0.6 ms
11	PMS-F-008 Fibonacci estimates › rejects invalid estimate undefined before replacing saved tasks	Pass	0.6 ms
12	PMS-F-008 Fibonacci estimates › rejects invalid estimate before replacing saved tasks *(input: an empty string)*	Pass	1.1 ms
13	PMS-F-008 Fibonacci estimates › rejects invalid estimate 5 before replacing saved tasks *(input: the text "5", not the number 5)*	Pass	1.0 ms
14	PMS-F-008 Fibonacci estimates › rejects invalid estimate true before replacing saved tasks	Pass	0.9 ms

#	Test case	Result	Time
15	PMS-F-008 Fibonacci estimates › persists every allowed estimate through a read after save	Pass	0.7 ms
16	PMS-F-008 Fibonacci estimates › keeps the seed data on the scale so a fresh board can be saved	Pass	2.5 ms

A.24 `supabaseSync.test.js` (15 tests, 15 passed)

#	Test case	Result	Time
1	PMS-F-015 Supabase synchronisation › reads project, sprint and task records from Supabase and keeps a local copy	Pass	10.3 ms
2	PMS-F-015 Supabase synchronisation › publishes local data in foreign-key order on first connection to an empty project	Pass	3.1 ms
3	PMS-F-015 Supabase synchronisation › maps rows onto columns that really exist in database/schema.sql	Pass	5.8 ms
4	PMS-F-015 Supabase synchronisation › upserts only those columns, dropping client-only fields	Pass	2.0 ms
5	PMS-F-015 Supabase synchronisation › stores a change and its audit record locally before the request, even if it never answers	Pass	2.7 ms
6	PMS-F-015 Supabase synchronisation › repairs the 6-point task left by earlier seed data so the board can be saved	Pass	2.3 ms
7	PMS-F-015 Supabase synchronisation › names the task when a stored estimate cannot be saved	Pass	1.9 ms
8	PMS-F-015 Supabase synchronisation › surfaces a sync error without discarding the local change	Pass	1.2 ms
9	PMS-F-015 Supabase synchronisation › deletes the task row in Supabase so a deleted task cannot come back	Pass	0.7 ms
10	PMS-F-015 Supabase synchronisation › stores audit records in activity_logs and never overwrites or duplicates one	Pass	1.4 ms
11	PMS-F-015 Supabase synchronisation › loads project audit records from Supabase and caches the mapped history	Pass	1.0 ms
12	PMS-F-015 Supabase synchronisation › merges local-only audit events with cloud history without duplicates	Pass	0.9 ms
13	PMS-F-015 Supabase synchronisation › falls back to local audit history and reports a cloud read failure	Pass	0.9 ms
14	PMS-F-015 Supabase synchronisation › TC-SET-01: rejects a malformed URL and keeps the previous configuration	Pass	1.5 ms
15	PMS-F-015 Supabase synchronisation › seeds UUID identifiers that satisfy the schema and its foreign keys	Pass	1.7 ms

A.25 `taskExplorer.test.js` (10 tests, 10 passed)

#	Test case	Result	Time
1	Task explorer and CSV › combines every predicate without mutating input	Pass	3.4 ms
2	Task explorer and CSV › orders priorities, estimates and titles	Pass	14.0 ms
3	Task explorer and CSV › neutralises spreadsheet formulas =1+1	Pass	0.5 ms
4	Task explorer and CSV › neutralises spreadsheet formulas +cmd	Pass	0.2 ms
5	Task explorer and CSV › neutralises spreadsheet formulas -cmd	Pass	0.1 ms
6	Task explorer and CSV › neutralises spreadsheet formulas @SUM(1)	Pass	0.1 ms
7	Task explorer and CSV › neutralises spreadsheet formulas =1 *(input: "=1" after two spaces)*	Pass	0.2 ms
8	Task explorer and CSV › neutralises spreadsheet formulas ⇥=cmd *(input: "=cmd" after a tab)*	Pass	0.2 ms
9	Task explorer and CSV › escapes quotes, delimiters, unicode and newlines	Pass	0.6 ms
10	Task explorer and CSV › exports an empty table with headers	Pass	0.3 ms

A.26 `taskFilters.test.js` (2 tests, 2 passed)

#	Test case	Result	Time
1	task board filters › combines assignee, priority, and text query using every selected predicate	Pass	3.2 ms
2	task board filters › reset clears assignee, priority, and search state	Pass	2.5 ms

A.27 `taskForm.test.js` (5 tests, 5 passed)

#	Test case	Result	Time
1	Task form validation and persistence › requires a nonblank title and a supported priority	Pass	4.6 ms
2	Task form validation and persistence › creates and persists every task form field	Pass	3.5 ms
3	Task form validation and persistence › edits and persists task fields while preserving task identity	Pass	1.1 ms
4	Task form validation and persistence › restores persisted values when reopening an edit form	Pass	0.4 ms
5	Task form validation and persistence › preserves backlog and unassigned values when an existing task is saved	Pass	0.3 ms

A.28 `taskSignals.test.js` (1 test, 1 passed)

#	Test case	Result	Time
1	pairs non-colour labels and icons for every priority and lifecycle status	Pass	7.4 ms

A.29 `taskUpdates.test.js` (3 tests, 3 passed)

#	Test case	Result	Time
1	Kanban task transitions › moves a dropped task to the target stage and ignores a same-stage drop	Pass	3.6 ms
2	Kanban task transitions › records the local update duration	Pass	4.3 ms
3	Kanban task transitions › records actor, task, prior and next stage, and timestamp for the transition	Pass	0.7 ms

A.30 `userText.test.js` (4 tests, 4 passed)

#	Test case	Result	Time
1	shared user text sanitization › neutralizes script and HTML payloads in task title, description, and tags before saving	Pass	5.4 ms
2	shared user text sanitization › neutralizes project name and description and enforces their length limits	Pass	5.1 ms
3	shared user text sanitization › cleans tag and search-view text, while React renders tags as text	Pass	7.7 ms
4	shared user text sanitization › enforces task title, description, and tag limits and rejects a title sanitized to blank	Pass	0.8 ms

A.31 `workflow.test.js` (8 tests, 8 passed)

#	Test case	Result	Time
1	Workflow Enums and Constants Verification › verifies 5 Kanban status columns are defined in sequence	Pass	3.1 ms
2	Workflow Enums and Constants Verification › provides all five named columns in board order	Pass	0.7 ms
3	Workflow Enums and Constants Verification › counts exactly the tasks shown in each matching column	Pass	0.8 ms
4	Workflow Enums and Constants Verification › verifies Team 11 members match official PES allocation	Pass	1.2 ms
5	Workflow Enums and Constants Verification › verifies Fibonacci estimation scale	Pass	0.3 ms
6	Workflow Enums and Constants Verification › advances a task by exactly one stage with one update call	Pass	2.4 ms
7	Workflow Enums and Constants Verification › has no enabled next-stage action for Done tasks	Pass	0.6 ms
8	Workflow Enums and Constants Verification › persists an advanced task through the local data service	Pass	1.1 ms

A.32 `workload.test.js` (5 tests, 5 passed)

#	Test case	Result	Time
1	PMS-F-013 team workload and capacity › matches each member's assigned and completed points to their own tasks	Pass	4.2 ms
2	PMS-F-013 team workload and capacity › reports a 0% completion rate, never NaN, for a member with no assigned points	Pass	0.6 ms
3	PMS-F-013 team workload and capacity › reconciles member totals with the sprint total by separating unowned work	Pass	1.1 ms
4	PMS-F-013 team workload and capacity › recalculates after a task is reassigned or completed	Pass	0.5 ms

#	Test case	Result	Time
5	PMS-F-013 team workload and capacity › renders every member with their points and the unowned work	Pass	10.4 ms

A.33 `workSignals.test.js` (3 tests, 3 passed)

#	Test case	Result	Time
1	validates real calendar dates and sanitizes bounded blocker text	Pass	7.7 ms
2	classifies overdue and blocked work while excluding completed work	Pass	0.8 ms
3	audits planning changes and protects exported blocker formulas	Pass	4.4 ms

A.34 `workspaceFeedback.test.js` (5 tests, 5 passed)

#	Test case	Result	Time
1	defaults to visual motion with opt-in vibration and handles corrupt preferences	Pass	3.6 ms
2	never vibrates without consent, activation, capability, or with reduced motion	Pass	3.2 ms
3	single-key actions never interrupt typing, composition or platform shortcuts	Pass	0.7 ms
4	validates appearance preferences and preserves legacy settings	Pass	0.4 ms
5	keeps preference changes usable when browser storage is denied	Pass	3.9 ms

A.35 `workViews.test.js` (2 tests, 2 passed)

#	Test case	Result	Time
1	combines a selected scope with distinct actionable quick views	Pass	4.3 ms
2	keeps empty scopes empty and does not assign someone else’s tasks to the current user	Pass	0.5 ms

Appendix B: Reproducing These Results

```
npm ci
npm test
# raw results in the same formats as this report
npx vitest run --reporter=default --reporter=junit --reporter=json   --outputFile.junit=unit-test-results.junit.xml -
```